Chapter 31: Money Growth and Inflation

If printing money can make people feel richer in the short run, why do economies that try it often end up poorer in the long run?

Learning Objectives

31.1 Explain the quantity theory of money and its implications for inflation.
31.2 Differentiate between nominal and real variables in the classical dichotomy.
31.3 Apply the Fisher effect to link nominal interest rates, real interest rates, and inflation.
31.4 Identify and evaluate the economic costs of inflation.
31.5 Analyze policy and real-world implications of inflation and deflation.

Key Points

- The overall price level adjusts to balance the money supply and money demand. When the central bank expands the money supply, prices rise; persistent money growth leads to ongoing inflation.
- In the long run, money is neutral—it affects nominal variables like prices and wages but not real variables such as output or employment.
- Governments that finance spending by printing money impose an inflation tax, and heavy reliance on it can cause hyperinflation.
- The Fisher effect links inflation and interest rates: as expected inflation rises, nominal interest rates increase while real rates remain unchanged.
- Inflation does not directly make people poorer, since it raises both prices and incomes, but it still imposes economic costs.
- The main costs of inflation include:
    - Shoeleather and menu costs from managing money and changing prices.
    - Distorted relative prices and tax burdens due to inflation's uneven effects.
    - Confusion and wealth redistribution between debtors and creditors, especially during high inflation.

The Value of Money

- P = the price level (e.g., the CPI or GDP deflator) is the price of a basket of goods, measured in money.
- $1/P$ is the value of $1, measured in goods.
- Example: basket contains one candy bar.
    - If P = $2, value of $1 is 1/2 candy bar
    - If P = $3, value of $1 is 1/3 candy bar
- Inflation drives up prices and drives down the value of money.

Money Supply (MS)

- In real world, determined by Federal Reserve, the banking system, consumers.

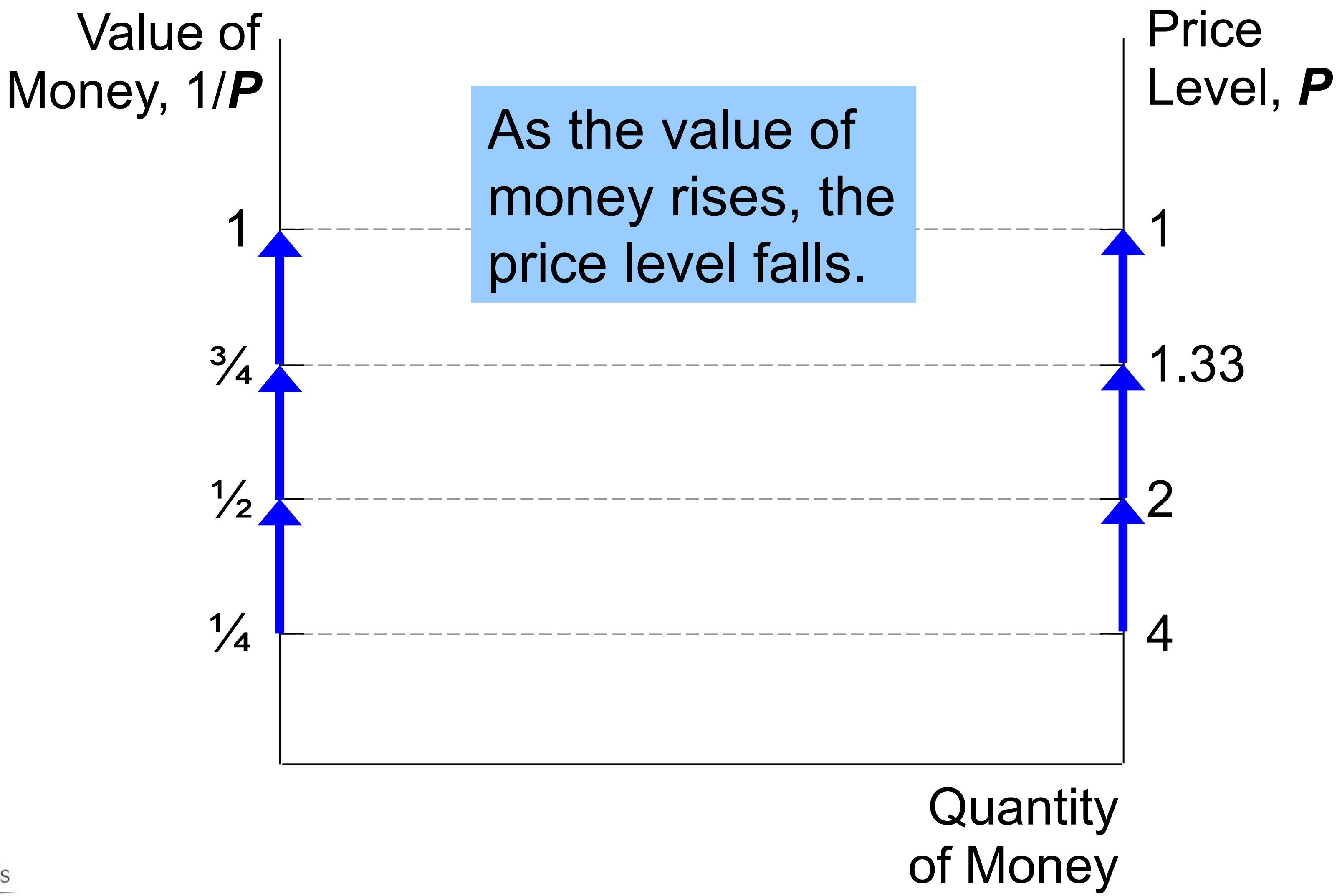
- In this model, we assume the Fed precisely controls MS and sets it at some fixed amount.

Money Demand (MD)

- Refers to how much wealth people want to hold in liquid form.
- Depends on ***P*** : An increase in ***P*** reduces the value of money, so, more money is required to buy goods and services.
- Thus, quantity of money demanded is negatively related to the value of money and positively related to ***P***, other things equal.

  (These "other things" include real income, interest rates, availability of ATMs.)

The Money Supply-Demand Diagram



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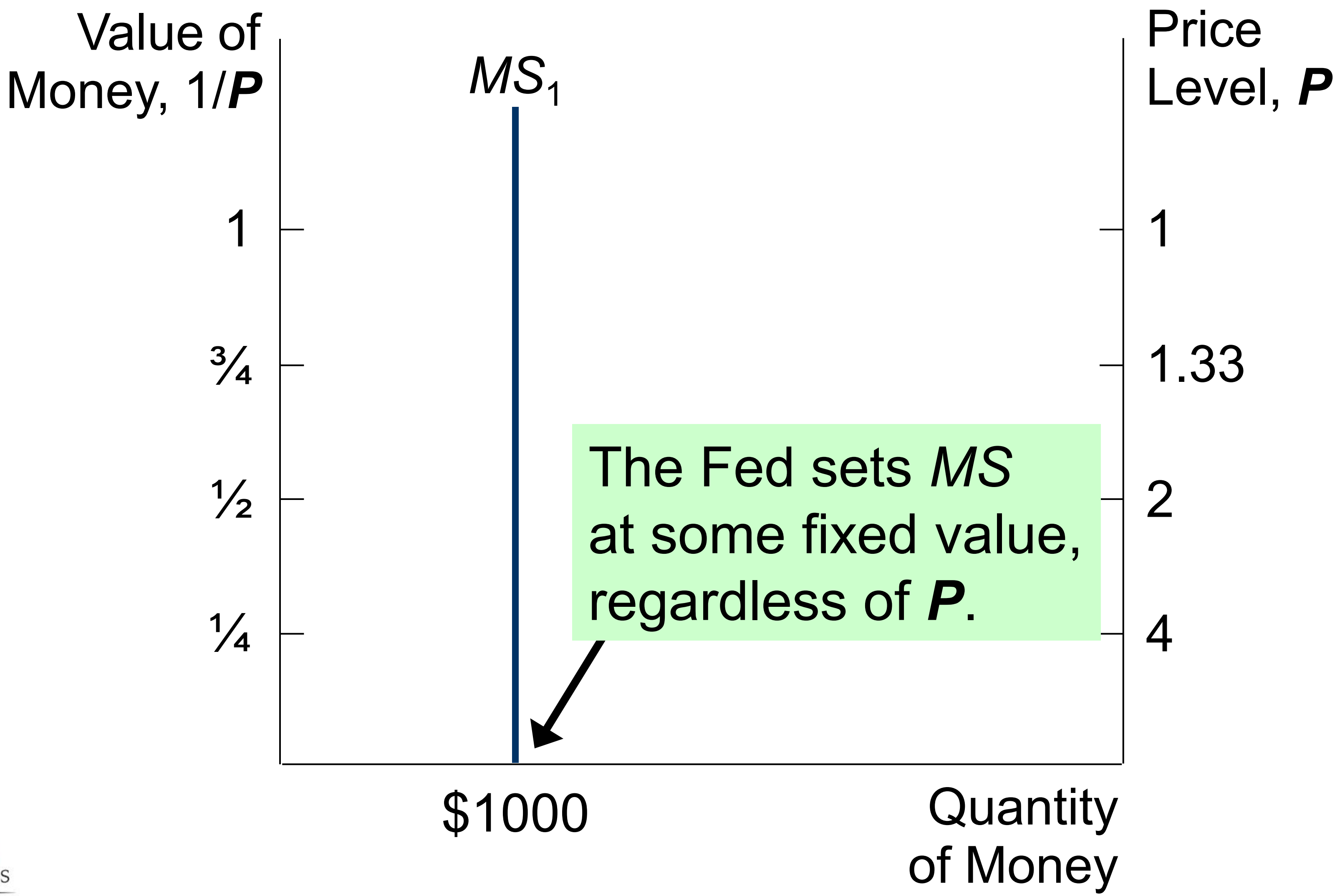
The Money Supply-Demand Diagram



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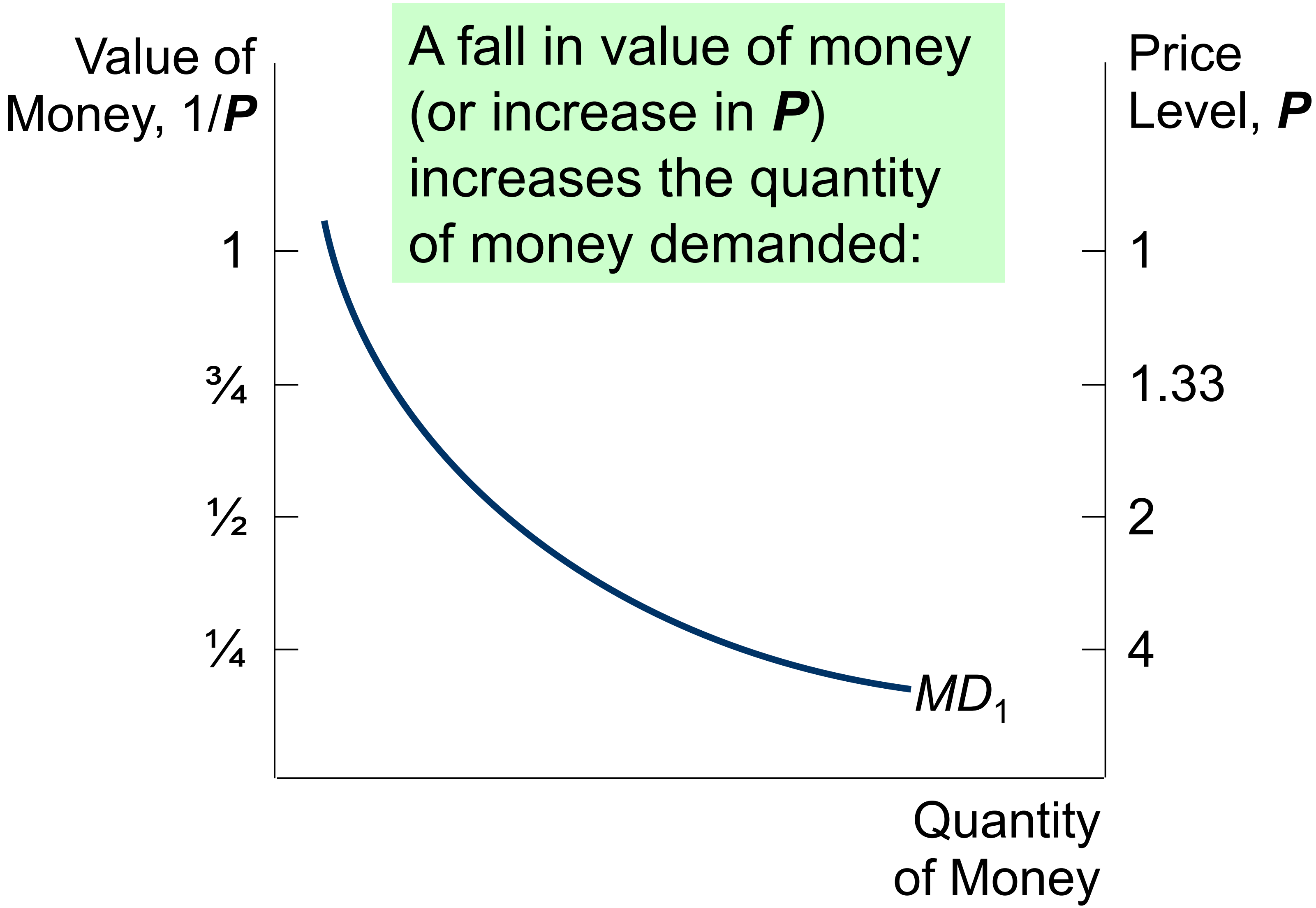
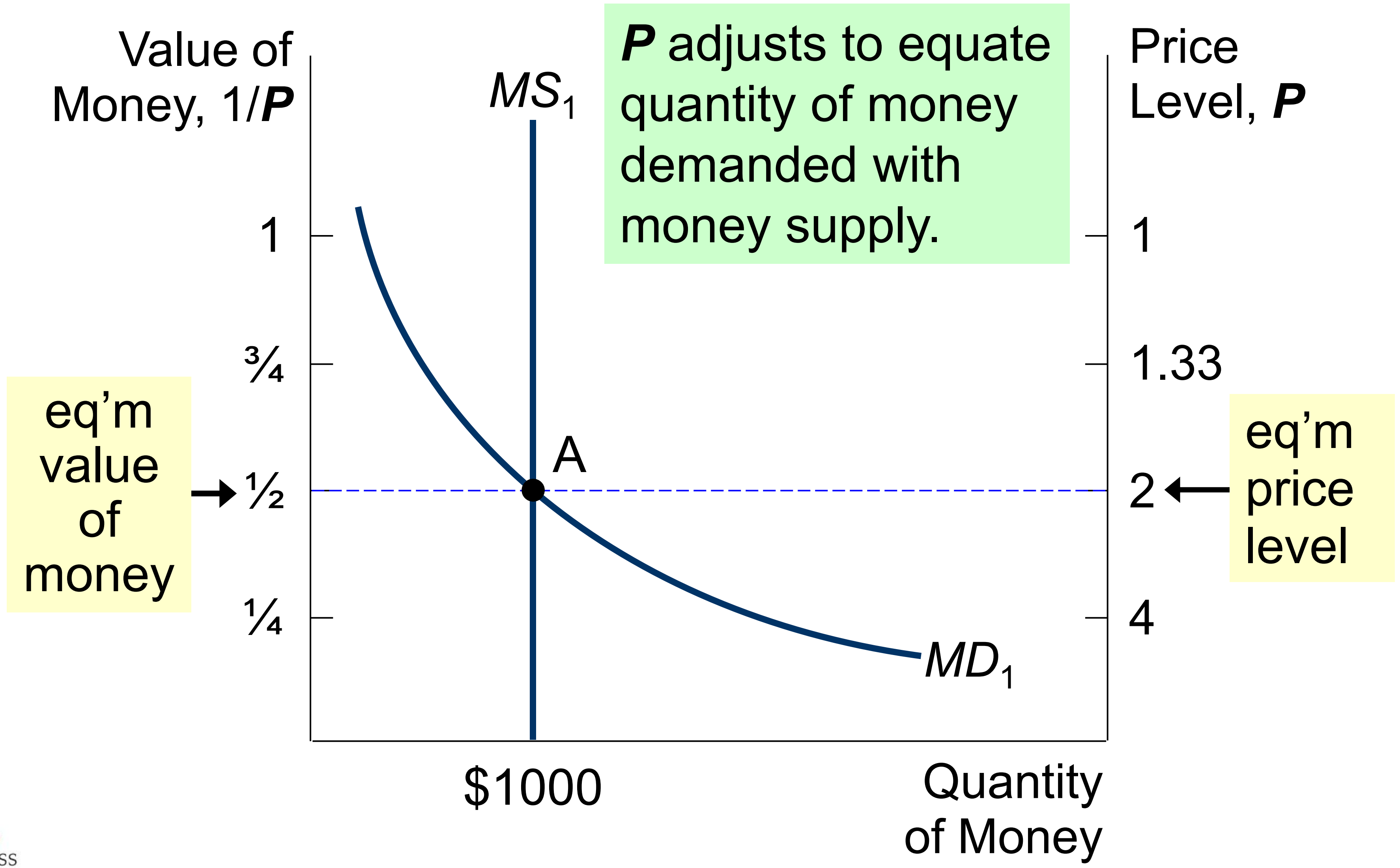


The Money Supply-Demand Diagram



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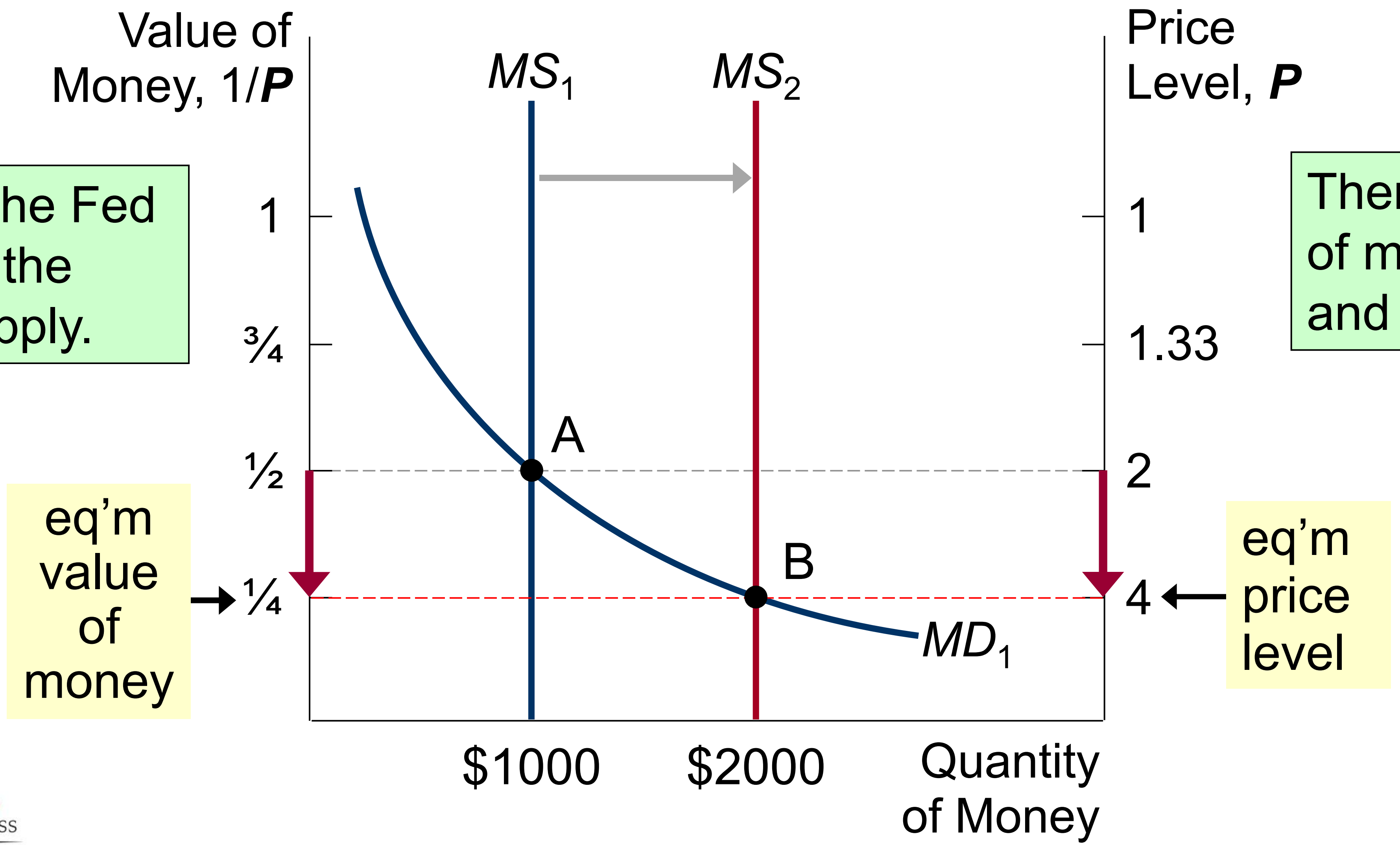
The Effects of a Monetary Injection

Suppose the Fed increases the money supply.



Then the value of money falls, and ***P*** rises.

A Brief Look at the Adjustment Process

- Result from graph: Increasing MS causes ***P*** to rise.
- How does this work? Short version:
  - At the initial ***P***, an increase in MS causes excess supply of money.
  - People get rid of their excess money by spending it on goods & services
  - or by loaning it to others, who spend it. Result: increased demand for goods.
  - But supply of goods does not increase, so prices must rise.
- (Other things happen in the short run, which we will study in later chapters.)

Real vs. Nominal Variables

- **Nominal variables** are measured in monetary units.

  Examples: nominal GDP, nominal interest rate (rate of return measured in $), nominal wage ($ per hour worked)

- **Real variables** are measured in physical units.

  Examples: real GDP, real interest rate (measured in output), real wage (measured in output)

Real vs. Nominal Variables

Prices are normally measured in terms of money.

Price of a toilet paper: $16/tp

Price of a bleach wipes: $8/wipes

A **relative price** is the price of one good relative to (divided by) another:

Relative price of TP in terms of wipes:

$$\frac{\text{price of tp}}{\text{price of wipes}} = \frac{\$16/\text{tp}}{\$8/\text{wipes}} = \boxed{\text{2 wipes per tp}}$$

Relative prices are measured in physical units, so they are real variables.

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Real vs Nominal Wage

An important relative price is the real wage:

W = nominal wage = price of labor, e.g., \$15/hour

P = price level = price of goods & services, e.g., \$5/unit of output

Real wage is the price of labor relative to the price of output:

$$\frac{\boldsymbol{W}}{\boldsymbol{P}} = \frac{\$15/\text{hour}}{\$5/\text{unit of output}} = 3 \text{ units output per hour}$$

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The Classical Dichotomy

- **Classical dichotomy**: the theoretical separation of nominal and real variables
- Hume and the classical economists suggested that monetary developments affect nominal variables but not real variables.
- If central bank doubles the money supply, Hume & classical thinkers contend
  - all nominal variables—including prices—will double.
  - all real variables—including relative prices—will remain unchanged.

The Neutrality of Money

- **Monetary neutrality**: the proposition that changes in the money supply do not affect real variables
- Doubling money supply causes all nominal prices to double; what happens to relative prices?
- Initially, relative price of toilet paper in terms of bleach wipes is

$$\frac{\text{price of tp}}{\text{price of wipes}} = \frac{\$16/\text{tp}}{\$8/\text{wipes}} = \underbrace{\text{2 wipes per tp}}$$

The relative price is unchanged.

<u>After nominal prices double,</u>

$$\frac{\text{price of tp}}{\text{price of wipes}} = \frac{\$32/\text{tp}}{\$16/\text{wipes}} = \overbrace{\text{2 wipes per tp}}$$

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The Neutrality of Money

Monetary neutrality: the proposition that changes in the money supply do not affect real variables

- Similarly, the real wage ***W*/*P*** remains unchanged, so
  - quantity of labor supplied does not change
  - quantity of labor demanded does not change
  - total employment of labor does not change
- The same applies to employment of capital and other resources.
- Since employment of all resources is unchanged, total output is also unchanged by the money supply.

The Velocity of Money

- **Velocity of money**: the rate at which money changes hands
- Notation:

$\boldsymbol{P} \text{ x } \boldsymbol{Y}$ = nominal GDP

= (price level) x (real GDP)

$\boldsymbol{M}$ = money supply

$\boldsymbol{V}$ = velocity

- Velocity formula:

$$\boldsymbol{V} = \frac{\boldsymbol{P} \text{ x } \boldsymbol{Y}}{\boldsymbol{M}}$$

The Velocity of Money

Velocity formula: $V = \dfrac{P \times Y}{M}$

Example with one good: bleach wipes.
In 2020,

Y = real GDP = 5,000 wipes

P = price level = price of wipes = $8

$P \times Y$ = nominal GDP = value of wipes = $40,000

M = money supply = $10,000

V = velocity = $40,000/$10,000 = 4

The average dollar was used in 4 transactions.

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Activity: NGDP and Velocity

One good: chicken.

The economy has enough labor, capital, and land to produce:

- $\boldsymbol{Y}$ = 5,000 pounds of chicken.
- $\boldsymbol{V}$ is constant.

In 2019, MS = \$4,000 and $\boldsymbol{P}$ = \$2/pound.

Compute nominal GDP and velocity in 2019.

Activity: NGDP and Velocity

Given: $\boldsymbol{Y} = 5{,}000$, $\boldsymbol{V}$ is constant,
MS = \$4,000 and $\boldsymbol{P} = \$2$ in 2019.

Compute nominal GDP and velocity in 2019.

$$\text{Nominal GDP} = \boldsymbol{P} \times \boldsymbol{Y} = \$2 \times 5000 = \boxed{\$10{,}000}$$

$$\boldsymbol{V} = \frac{\boldsymbol{P} \times \boldsymbol{Y}}{\boldsymbol{M}} = \frac{\$10{,}000}{\$4{,}000} = \boxed{2.5}$$

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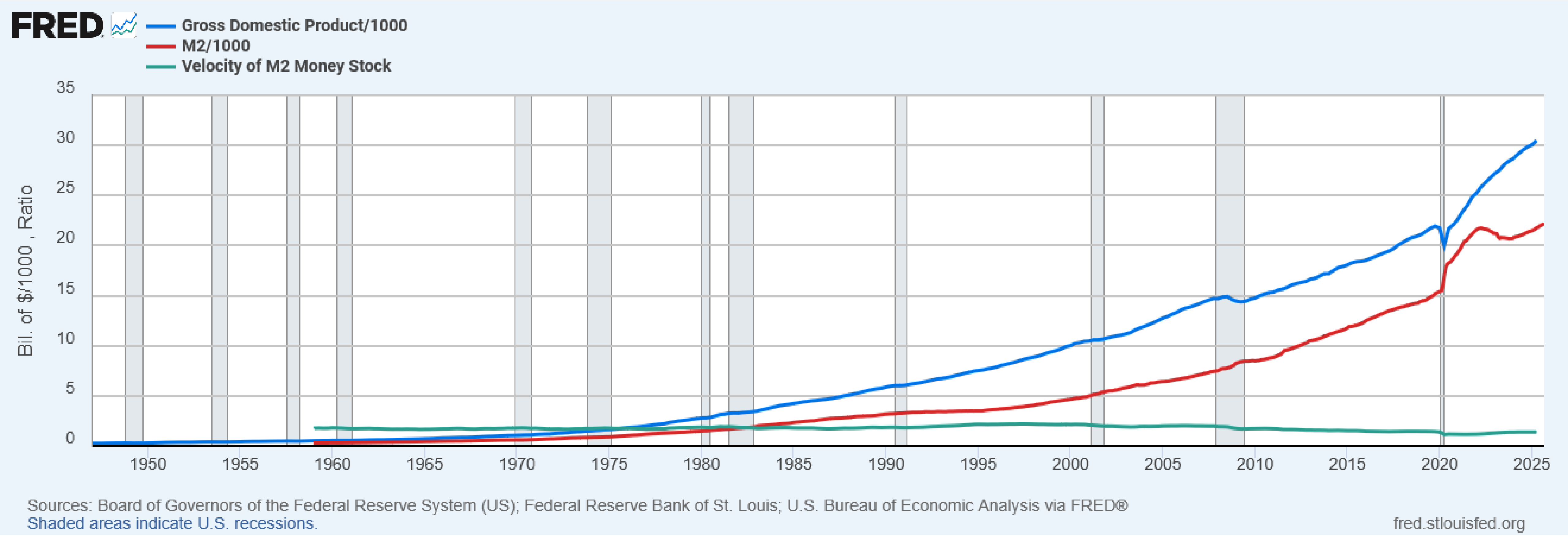
U.S. Nominal GDP, M2 and Velocity



The Quantity Equation

Velocity formula: $$V = \frac{P \text{ x } Y}{M}$$

- Multiply both sides of formula by M:

$$M \text{ x } V = P \text{ x } Y$$

- Called the **quantity equation**

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The Quantity Theory in 5 Steps

Start with quantity equation: $\boldsymbol{M} \text{ x } \boldsymbol{V} = \boldsymbol{P} \text{ x } \boldsymbol{Y}$

1. $\boldsymbol{V}$ is stable.
2. So, a change in $\boldsymbol{M}$ causes nominal GDP ($\boldsymbol{P}$ x $\boldsymbol{Y}$) to change by the same percentage.
3. A change in $\boldsymbol{M}$ does not affect $\boldsymbol{Y}$:
   money is neutral, $\boldsymbol{Y}$ is determined by technology & resources
4. So, $\boldsymbol{P}$ changes by same percentage as $\boldsymbol{P}$ x $\boldsymbol{Y}$ and $\boldsymbol{M}$.
5. Rapid money supply growth causes rapid inflation.

Activity: Inflation

One good: chicken. The economy has enough labor, capital, and land to produce $\boldsymbol{Y} = 5{,}000$ pounds of chicken. $\boldsymbol{V}$ is constant at 2.5. In 2019, MS = \$4,000, $\boldsymbol{P} = \$2/\text{pound}$.

For 2020, the Fed increases MS by 67%, to \$6,680.

a. Compute the 2020 values of nominal GDP and $\boldsymbol{P}$. Compute the inflation rate for 2019–2020.

b. Suppose tech. progress causes $\boldsymbol{Y}$ to increase to 5,250 in 2020. Compute 2019–2020 inflation rate.

Activity: Inflation

Given: $Y = 5{,}000$, V is constant at 2.5, MS = \$4,000 and P = \$2 in 2019.

For 2020, the Fed increases MS by 67%, to \$6,680.

a. Compute the 2020 values of nominal GDP and P. Compute the inflation rate for 2019–2020.

$$\text{Nominal GDP} = P \times Y = M \times V \quad \text{(Quantity Equation)}$$

$$= \$6{,}680 \times 2 = \boxed{\$16{,}700}$$

$$P = \frac{P \times Y}{Y} = \frac{\$16{,}700}{5{,}000} = \boxed{\$3.34}$$

$$\text{Inflation rate} = \frac{\$3.34 - 2.00}{2.00} = \boxed{67\%} \quad \text{(same as MS!)}$$

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Activity: Inflation

Given: $Y = 5{,}000$, V is constant at 2.5, MS = \$4,000 and P = \$2 in 2019.

For 2020, the Fed increases MS by 67%, to \$6,680.

b. Suppose tech. progress causes Y to increase 5% in 2020, to 5,250. Compute 2019–2020 inflation rate.

First, use Quantity Equation to compute P in 2020:

$$P = \frac{M \text{ x } V}{Y} = \frac{\$16{,}700}{5{,}250} = \$3.18$$

$$\text{Inflation rate} = \frac{\$3.18 - 2.00}{2.00} = \boxed{59\%}$$

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Summary and Lessons about the Quantity Theory of Money

- If real GDP is constant, then inflation rate = money growth rate.
- If real GDP is growing, then inflation rate < money growth rate.
- The bottom line:
  - Economic growth increases # of transactions.
  - Some money growth is needed for these extra transactions.
  - Excessive money growth causes inflation.

Hyperinflation

- Hyperinflation is generally defined as inflation exceeding 50% per month.
- Recall one of the Ten Principles from Chapter 1:
  Prices rise when the government prints too much money.
- Excessive growth in the money supply always causes hyperinflation.

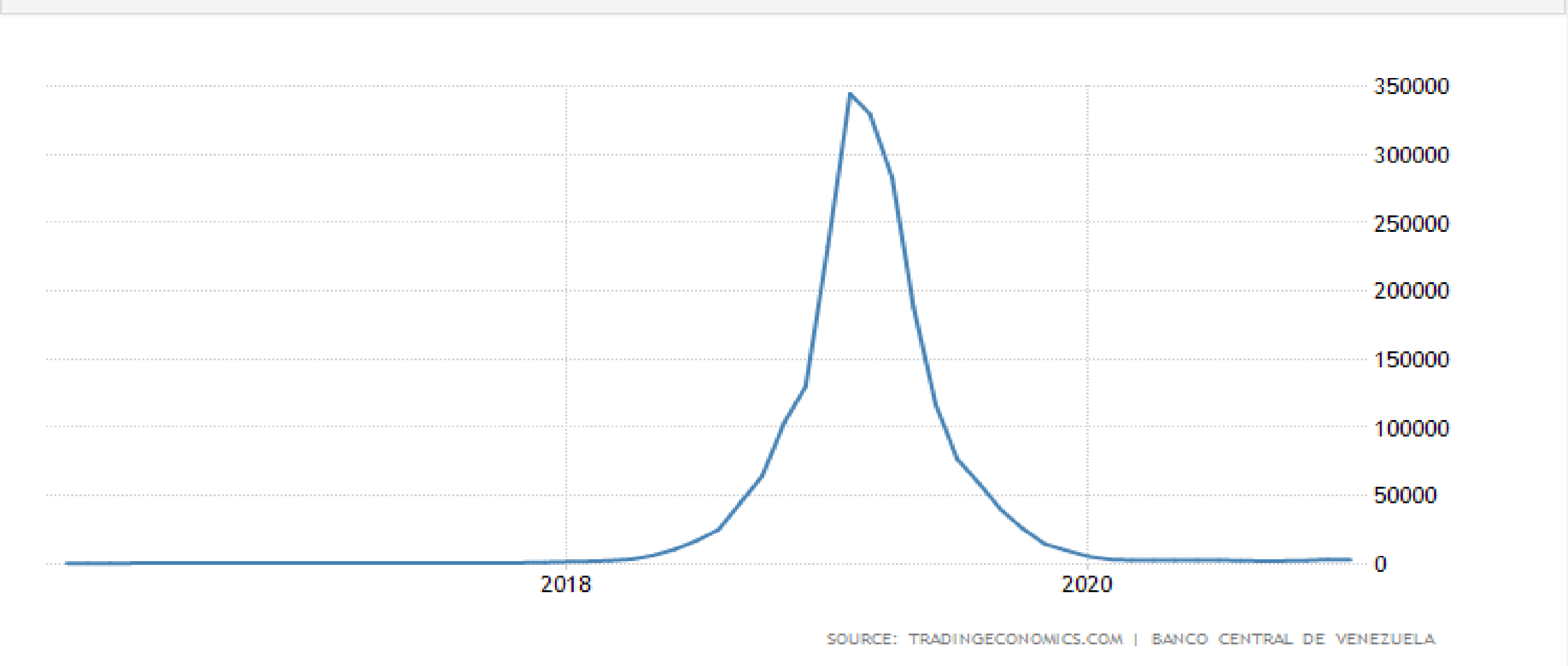
Hyperinflation in Zimbabwe

Large government budget deficits led to the creation of large quantities of money and high inflation rates.

Sign posted in public restroom

Date	Zim$ per US$
Aug 2007	245
Apr 2008	29,401
May 2008	207,209,688
June 2008	4,470,828,401
July 2008	26,421,447,043
Feb 2009	37,410,030
Sept 2009	355

Hyperinflation in Venezuela



Actual	Previous	Highest	Lowest	Dates	Unit	Frequency	
2665.40	2959.80	344509.50	3.22	1973 - 2021	percent	Monthly	2007=100

[Source: tradingeconomics.com](#)

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The Inflation Tax

- When tax revenue is inadequate and ability to borrow is limited, government may print money to pay for its spending.
- Almost all hyperinflations start this way.
- The revenue from printing money is the **inflation tax**: printing money causes inflation, which is like a tax on everyone who holds money.
- In the U.S., the inflation tax today accounts for less than 3% of total revenue.

The Fisher Effect

Rearrange the definition of the real interest rate:

$$\text{Nominal interest rate} = \text{Inflation rate} + \text{Real interest rate}$$

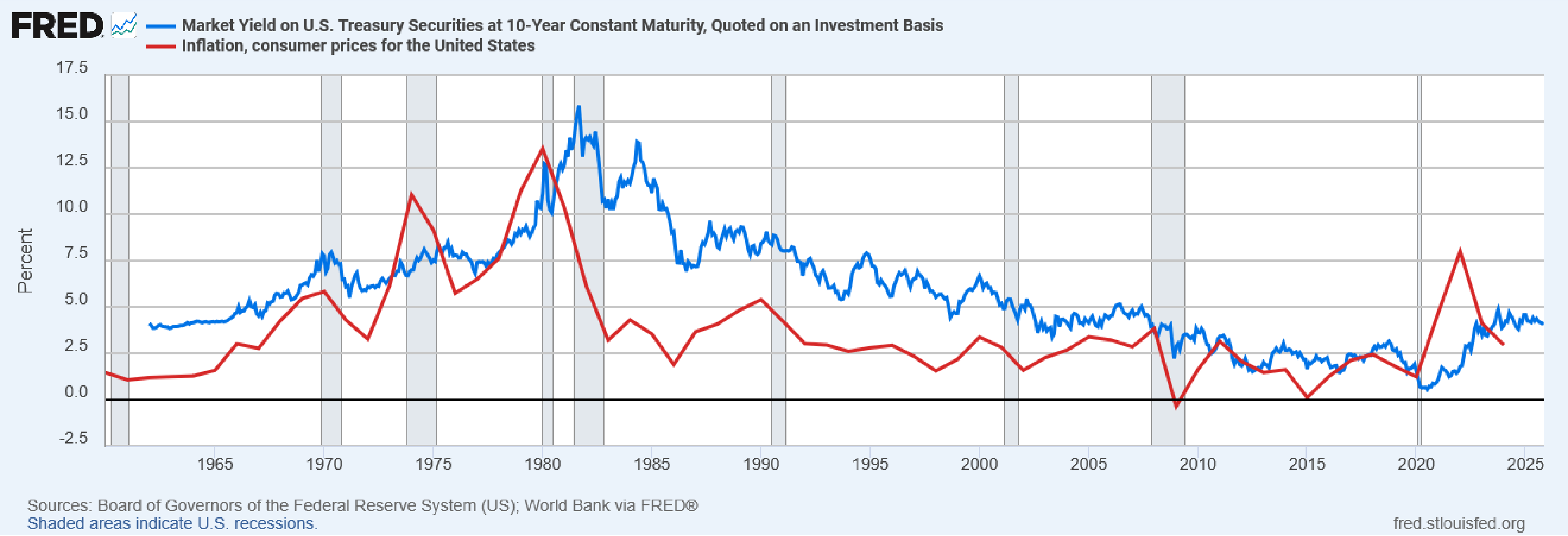
- The real interest rate is determined by saving & investment in the loanable funds market.
- Money supply growth determines inflation rate.
- So, this equation shows how the nominal interest rate is determined.

The Fisher Effect

Nominal interest rate	=	Inflation rate	+	Real interest rate

- In the long run, money is neutral, so a change in the money growth rate affects the inflation rate but not the real interest rate.
- So, the nominal interest rate adjusts one-for-one with changes in the inflation rate.
- This relationship is called the **Fisher effect** after Irving Fisher, who studied it.

US Nominal Interest and Inflation Rates

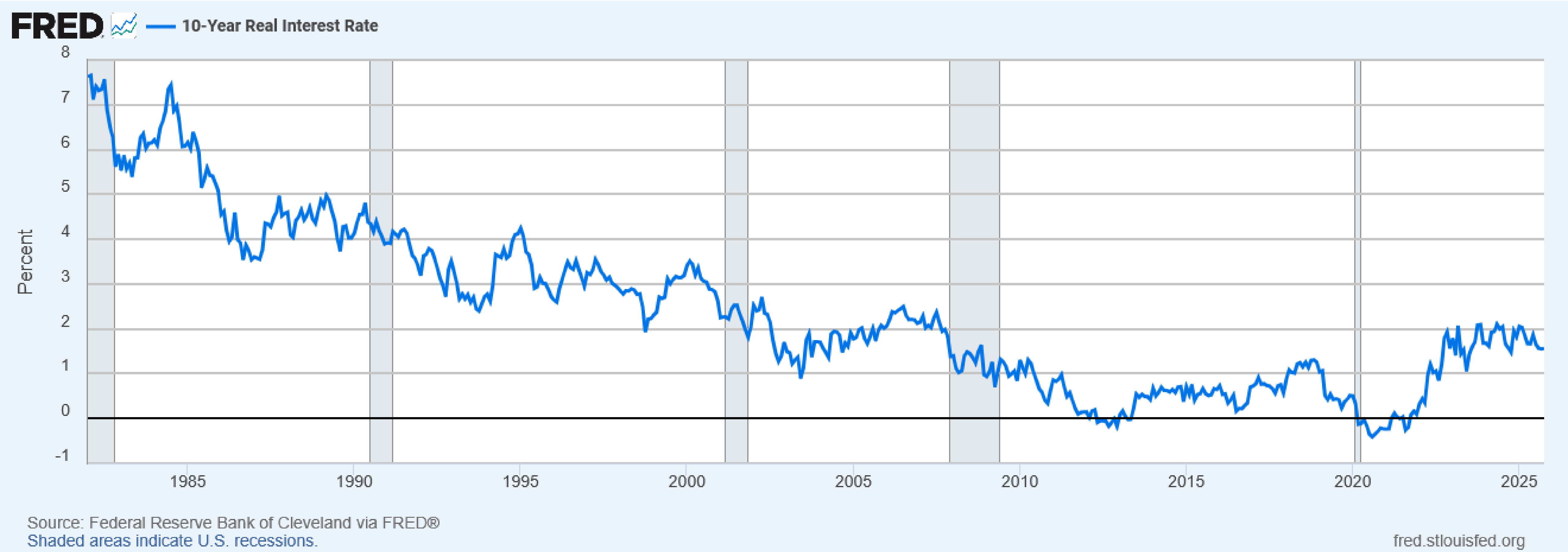


The Fisher Effect & the Inflation Tax

$$\text{Nominal interest rate} = \text{Inflation rate} + \text{Real interest rate}$$

- The inflation tax applies to people's holdings of money, not their holdings of wealth.
- The Fisher effect: an increase in inflation causes an equal increase in the nominal interest rate,
  so the real interest rate (on wealth) is unchanged.

10-Year Real Interest Rate



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Activity: The Fisher Effect in Action

Scenario:
The economy's real interest rate is determined by savings and investment at 3%.
The central bank increases the money growth rate, raising expected inflation from 2% to 6%.

Questions:

- What happens to the nominal interest rate according to the Fisher effect?
- What happens to the real interest rate in the long run?
- If a borrower agreed to a fixed-rate loan before inflation expectations changed, who gains and who loses?
- Discuss: How might policymakers use this relationship when inflation expectations shift suddenly?

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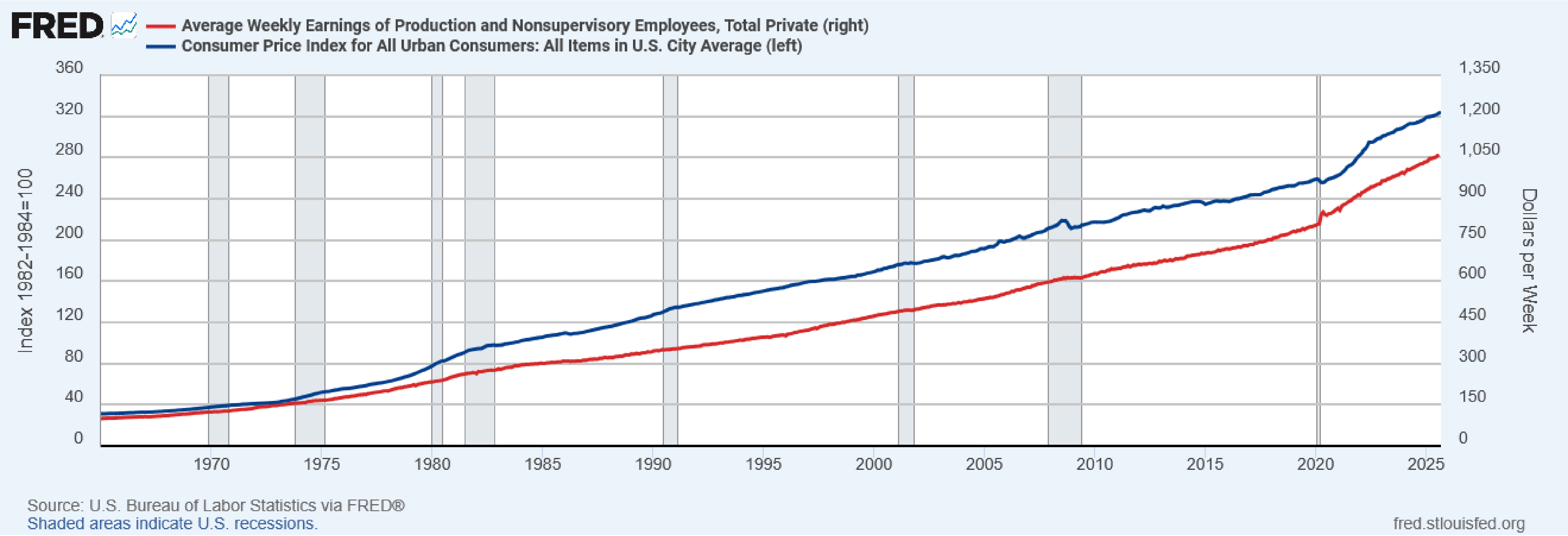
Activity: The Fisher Effect in Action

1. Nominal interest rate = Real + Expected inflation = 3% + 6% = 9%.
2. The real rate remains 3% in the long run (money neutrality).
3. Borrowers gain because they repay loans with less valuable dollars; lenders lose.
4. Policymakers must anticipate inflation expectations — if they rise, nominal rates must increase to maintain real returns

The Costs of Inflation

- The **inflation fallacy**: most people think inflation erodes real incomes.
- But inflation is a general increase in prices of the things people buy and the things they sell (e.g., their labor).
- In the long run, real incomes are determined by real variables, not the inflation rate.
- What about the short run?

US Average Hourly Earnings & the CPI



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The Costs of Inflation

- **Shoe leather costs**: the resources wasted when inflation encourages people to reduce their money holdings
  - Includes the time and transactions costs of more frequent bank withdrawals
- **Menu costs**: the costs of changing prices
  - Printing new menus, mailing new catalogs, etc.
- **Misallocation of resources from relative-price variability**: Firms don't all raise prices at the same time, so relative prices can vary, which distorts the allocation of resources.
- **Confusion & inconvenience**: Inflation changes the yardstick we use to measure transactions.
  - Complicates long-range planning and the comparison of dollar amounts over time.

The Costs of Inflation

- **Tax distortions**:
- Inflation makes nominal income grow faster than real income.
- Taxes are based on nominal income, and some are not adjusted for inflation.
- So, inflation causes people to pay more taxes even when their real incomes don’t increase.

A Special Cost of Unexpected Inflation

- **Arbitrary redistributions of wealth**
- Higher-than-expected inflation transfers purchasing power from creditors to debtors: Debtors get to repay their debt with dollars that aren't worth as much.
- ex: $1000 owed at 5% interest. With 0% inflation $1,050 is owed at year's end.
  - If unexpected inflation of 3% occurs, then at year's end the original $1,050 is only worth $1,020.
- Lower-than-expected inflation transfers purchasing power from debtors to creditors.
- High inflation is more variable and less predictable than low inflation.
- So, these arbitrary redistributions are frequent when inflation is high.

The Costs of Inflation

- All these costs are quite high for economies experiencing hyperinflation.
- For economies with low inflation (< 10% per year), these costs are probably much smaller, though their exact size is open to debate.

Key Takeaways

- The overall level of prices adjusts to balance money supply and money demand.
- When the central bank expands the money supply, prices rise; persistent money growth leads to continuing inflation.
- In the long run, money is neutral—it affects nominal variables (prices, wages, interest rates) but not real variables (output, employment).
- Governments that finance spending by printing money impose an inflation tax, and excessive reliance on it can lead to hyperinflation.
- The Fisher effect shows that when expected inflation rises, nominal interest rates increase while real rates remain unchanged.
- Inflation does not directly make people poorer since both prices and incomes rise, but it still creates economic inefficiencies.
- The main costs of inflation include:
  - Shoeleather costs from managing money holdings.
  - Menu costs from frequently changing prices.
  - Tax distortions and relative-price variability that misallocate resources.
  - Confusion and wealth redistribution between borrowers and lenders, especially during high inflation.
- Moderate inflation has small but real costs, while high or unpredictable inflation undermines confidence in money and economic stability.

Knowledge Check

1. Why does persistent money-supply growth cause sustained inflation?
2. If money supply grows 8 % and real GDP grows 3 %, what is the approximate inflation rate predicted by the quantity theory?
3. Explain how an unexpected increase in inflation affects borrowers and lenders.
4. If nominal interest rate = 9 % and expected inflation = 4 %, what is the real interest rate?

Knowledge Check Answers

1. Because faster money growth increases spending while real output is fixed in the long run, prices must rise.
2. Inflation ≈ Money growth – Real GDP growth = 8 – 3 = 5 %.
3. Borrowers repay loans with dollars that are worth less than expected, so borrowers gain and lenders lose purchasing power.
4. Real rate = Nominal – Expected Inflation = 9 – 4 **= 5 %.**